

From the Deterministic Regulator Data to a Gauge-Fixed Path Integral for the Finite $SU(3)$ Yang–Mills Lattice Theory

Abstract

The deterministic regulator paper constructs, for each regulator size N , the gauge-geometric part of a finite regularized $SU(3)$ data package

$$\mathcal{Q}^{(N)} = (\mathcal{N}^{(N)}, \mathcal{E}^{(N)}, \mathcal{C}_{\square}^{(N)}, \{U_3^{(N)}(e)\}_{e \in \mathcal{E}^{(N)}}, S_W^{(N)})$$

whose regulator geometry, sampled links, and Wilson action are fixed in [?]. The paper completes the finite-dimensional data package by adjoining the explicit fermionic and scalar actions written below and performs the quantization step: it replaces the sampled background links by independent integration variables and builds the exact finite-dimensional Euclidean path integral on that same regulator lattice. We define the full configuration space, the product Haar–Berezin–Lebesgue measure, the partition function, and the vacuum expectation map. We prove that the resulting finite-lattice theory is well defined for the free massive scalar sector used in this paper. We then solve the gauge-redundancy problem exactly by a rooted maximal-tree gauge: every gauge orbit meets the gauge slice once and only once, the Faddeev–Popov determinant is identically 1, and the ghost sector is a decoupled finite Berezin integral. The fermionic sector integrates to an exact finite fermion determinant, and the scalar sector is shown to be exactly Gaussian with an explicit lattice covariance.

The paper also establishes the Lie-algebraic structure of the discrete measure. On each edge, Haar measure is smooth in exponential coordinates and agrees to second order with Lebesgue measure on the Lie algebra. The asymptotic regime relevant to this programme is the stationary thermodynamic limit

$$\pi_{N,1} \Rightarrow \pi_{\infty},$$

already proved in the companion convergence paper. The convergence paper [?] proves this stationary mean-field limit, while [?] identifies the deterministic finite Wilson data and the continuum functional S_{YM}^{cont} on π_{∞} . The same convergence paper also proves the mean-field logarithmic Sobolev inequality on π_{∞} ; by tensorization, the same constant controls the finite product measures $\pi_{\infty}^{\otimes k}$ that support the thermodynamic observables. The paper supplies the exact fixed- N quantization, isolates a canonical continuum-indexed class of local gauge-invariant plaquette observables together with its finite-regulator approximation map, identifies the corresponding exact finite unnormalized Euclidean functionals, and singles out a distinguished localized Wilson sector whose sampled deterministic continuum limit is already identified in [?]. The transferred equilibrium stiffness continues to control the product measures supporting the thermodynamic plaquette presentation.

Contents

1 Introduction and motivation

From data to dynamics. The companion finite-regulator paper *Local Continuum Yang–Mills Limit from Particle Trajectories* constructs, for each regulator size, a finite deterministic spacetime lattice

$$\mathcal{L} = (\mathcal{N}, \mathcal{E})$$

whose site set and edge set form a deterministic coordinate graph, and whose canonical plaquette family $\mathcal{C}_\square$ is the corresponding plaquette set from [?]. On that lattice the same paper defines sampled link transporters $U_3^{(N)}(e) \in SU(3)$ and the Wilson action S_W . The paper places on the same regulator graph the finite matter actions S_F and S_ϕ defined in Definition ???. Together these objects form the finite data package $\mathcal{Q}^{(N)}$. Up to that stage, however, the link variables are deterministic functions of the continuum-seeded profile. A quantum theory requires a different mathematical object: instead of one distinguished configuration, it requires an ensemble weighted by an action and summed over all admissible gauge fields and matter fields. The paper performs that passage from one explicit background configuration to a finite-dimensional path integral on the same regulator lattice.

The quantization objective. The basic quantization step is conceptually simple and mathematically precise. The background transporters $U_3^{(N)}(e)$ derived in [?] are replaced by independent variables

$$U(e) \in SU(3), \quad e \in \mathcal{E},$$

and the site colors $\mathbf{c}_x$ are retained, when desired, as auxiliary fluctuating variables in the unit ball of $\mathbb{C}^3$. Because the companion paper derives no autonomous color-field energy beyond the induced links, the auxiliary color sector will be normalized and dynamically inert in the quantization performed here. All dynamical content therefore sits in the gauge links, fermions, and scalars. The goal is to define a mathematically complete finite partition function, remove gauge redundancy exactly, identify the fermion determinant and scalar measure, and clarify in what sense the resulting finite theory organizes the canonical local gauge-invariant observables used in the thermodynamic continuum programme.

Theorem 1.1 (Main finite-lattice quantization result). *Fix the lattice data $(\mathcal{N}, \mathcal{E}, \mathcal{C}_\square)$ from [?], the coupling $g_3 > 0$, masses $m_F > 0$ and $m_\phi > 0$, and the distance regularization $\eta > 0$. Then:*

1. *the promoted field manifold*

$$\mathcal{X}_\mathcal{L} := SU(3)^\mathcal{E} \times \mathbb{B}^\mathcal{N} \times \Lambda_F \times \mathbb{R}^\mathcal{N}$$

admits a canonical product measure obtained from normalized Haar measure on $SU(3)$, normalized Lebesgue measure on the color ball $\mathbb{B} \subset \mathbb{C}^3$, finite Berezin measure in the fermionic directions, and Lebesgue measure in the scalar directions;

2. *the partition function*

$$Z = \int_{\mathcal{X}_\mathcal{L}} \exp(-S_W(U) - S_F(U, \bar{\psi}, \psi) - S_\phi(\phi)) \, d\mu$$

exists as a finite complex number;

3. for any rooted spanning forest $\mathcal{T}$ of the underlying undirected graph, every rooted gauge orbit meets the maximal-tree gauge slice exactly once, the Faddeev–Popov determinant is identically 1, and the gauge-fixed partition function is an ordinary finite-dimensional integral over the cotree links;
4. integrating out the fermions yields the exact fermion determinant $\det M_F(U)$, while the scalar integral is exactly Gaussian with covariance $(L_\eta + m_\phi^2 I)^{-1}$.

Remark 1.2 (Scope of the paper). Theorem ?? is a complete quantization theorem for a *finite lattice*. The proofs below use only the finite graph

$$(\mathcal{N}, \mathcal{E}, \mathcal{C}_\square)$$

and the finite-dimensional gauge and matter actions fixed in the present construction. The measure-theoretic statements therefore concern exact finite-dimensional integrals, while the Lie-algebraic discussion in later sections concerns only local coordinate descriptions of Haar measure near the identity. In the broader series, these finite- N quantizations are then passed to the stationary thermodynamic limit. More precisely, [?] proves the weak convergence

$$\pi_{N,1} \Rightarrow \pi_\infty,$$

while [?] formulates the corresponding regulator finite action and continuum limit picture on π_∞ . The same mean-field law carries the logarithmic Sobolev inequality proved in [?], and Proposition ?? below makes explicit how that stiffness transfers to the finite product measures $\pi_\infty^{\otimes k}$ supporting the thermodynamic finite-lattice observables. The continuum-indexed observable class used for that thermodynamic presentation is fixed next. In addition, the paper records the distinguished localized Wilson sector whose sampled deterministic representatives are exactly the localized Wilson functionals of [?]; this is the direct bridge from the exact finite regulator theory constructed here to the continuum Yang–Mills functional identified in that companion paper.

Definition 1.3 (Continuum-indexed local gauge-invariant plaquette algebra). *Fix an orientation $0 \leq \mu < \nu \leq 3$, a compactly supported test function $\chi \in C_c(\mathbb{R}^4)$, and a bounded continuous class function $\Phi : SU(3) \rightarrow \mathbb{C}$, i.e.*

$$\Phi(VUV^{-1}) = \Phi(U) \quad \forall U, V \in SU(3).$$

The corresponding continuum-indexed local gauge-invariant plaquette observable is denoted by

$$\mathcal{O}_{\chi, \mu\nu, \Phi}.$$

Let $\mathfrak{A}_{\text{loc}}^{\text{g.i.}}$ be the unital algebra generated by finite linear combinations and finite products of such symbols.

Definition 1.4 (Canonical finite-regulator approximation map). *For each regulator size N , use the deterministic flat regulator data of [?]: the lattice spacing a_N , the plaquette family $\mathcal{C}_\square$, and the plaquette base-point map*

$$P \mapsto x_P \in \mathbb{R}^4.$$

For each orientation $0 \leq \mu < \nu \leq 3$, let $\mathcal{C}_{\square, \mu\nu} \subset \mathcal{C}_\square$ be the coordinate plaquettes of type (μ, ν) . Define

$$\iota_N(\mathcal{O}_{\chi, \mu\nu, \Phi})(U) := \sum_{P \in \mathcal{C}_{\square, \mu\nu}} a_N^4 \chi(x_P) \Phi(U(P)).$$

Extend ι_N linearly and multiplicatively to all of $\mathfrak{A}_{\text{loc}}^{\text{g.i.}}$.

Proposition 1.5 (Locality and boundedness of the canonical gauge-invariant observables). *The observable package of Definitions ?? and ?? has the following properties.*

1. For every $O \in \mathfrak{A}_{\text{loc}}^{\text{g.i.}}$, $\iota_N(O)$ is gauge invariant on the finite regulator lattice.
2. If O is generated by test functions $\chi_1, \dots, \chi_m$, then $\iota_N(O)$ depends only on plaquettes whose base points lie in the compact set $\bigcup_{j=1}^m \chi_j$, and hence only on the link variables meeting that compact window.
3. For each generator $\mathcal{O}_{\chi, \mu\nu, \Phi}$, there exists $C_{\chi, \Phi} < \infty$, independent of N , such that

$$\sup_{U \in \mathcal{U}} |\iota_N(\mathcal{O}_{\chi, \mu\nu, \Phi})(U)| \leq C_{\chi, \Phi}.$$

4. For every $O \in \mathfrak{A}_{\text{loc}}^{\text{g.i.}}$, the unnormalized Euclidean functional

$$\int_{\mathcal{X}_{\mathcal{L}}} \iota_N(O) e^{-S} d\mu$$

is finite.

Proof. For a closed plaquette P , a local gauge transformation Ω sends $U(P)$ to $\Omega(x_P)U(P)\Omega(x_P)^{-1}$, so gauge invariance of each generator follows from the class-function property of Φ . Finite products and linear combinations preserve gauge invariance.

The locality statement is immediate from the definition of ι_N : only plaquettes with base point $x_P \in \chi_j$ can contribute to the generator associated with χ_j , and finite products involve only the union of the corresponding compact supports.

Because Φ is bounded and χ is compactly supported, the deterministic regulator geometry from [?] implies that the number of contributing (μ, ν) -plaquettes is $O(a_N^{-4})$ on every fixed compact window, while each term carries the weight a_N^4 . Hence the sum is uniformly bounded in N and U .

Finally, $\iota_N(O)$ is bounded on the bosonic configuration space, so finiteness of the displayed Euclidean functional is exactly the unnormalized-integrability statement of Theorem ??. $\square$

Corollary 1.6 (Nontrivial local gauge-invariant probe). *The algebra $\mathfrak{A}_{\text{loc}}^{\text{g.i.}}$ is nontrivial.*

Proof. Take any nonzero $\chi \in C_c(\mathbb{R}^4)$, any $0 \leq \mu < \nu \leq 3$, and

$$\Phi(U) := 1 - \frac{1}{3} \text{Re tr } U.$$

This Φ is a bounded continuous class function on $SU(3)$, and it is not constant. Therefore, for every regulator level N for which χ contains at least one (μ, ν) -plaquette base point, the representative $\iota_N(\mathcal{O}_{\chi, \mu\nu, \Phi})$ is a nonconstant gauge-invariant observable on the finite regulator. $\square$

Definition 1.7 (Distinguished localized Wilson family). *For $\chi \in C_c(\mathbb{R}^4)$, let $\mathcal{W}_\chi$ denote the corresponding continuum-indexed localized Wilson symbol. Its canonical finite-regulator representative is the gauge-invariant plaquette sum*

$$j_N(\mathcal{W}_\chi)(U) := \frac{6}{g_3^2} \sum_{0 \leq \mu < \nu \leq 3} \sum_{P \in \mathcal{C}_{\square, \mu\nu}} \chi(x_P) \left(1 - \frac{1}{3} \text{Re tr } U(P) \right).$$

Theorem 1.8 (Canonical Wilson generators recover the sampled deterministic local continuum limit). *Let $\chi \in C_c(\mathbb{R}^4)$. For every regulator size N ,*

$$j_N(\mathcal{W}_\chi)(U) = \frac{6}{g_3^2} \sum_{0 \leq \mu < \nu \leq 3} \sum_{P \in \mathcal{C}_{\square, \mu\nu}} \chi(x_P) \left(1 - \frac{1}{3} \operatorname{Re} \operatorname{tr} U(P) \right).$$

In particular, when the gauge field U is specialized to the deterministic sampled plaquette holonomies $U_3^{(N)}(P)$ of [?], one has

$$j_N(\mathcal{W}_\chi)(U_3^{(N)}) = S_{\mathcal{W}, \chi}^{(N)},$$

the localized Wilson functional of the companion continuum-gauge paper. Consequently, the local continuum-limit theorem of [?] gives

$$j_N(\mathcal{W}_\chi)(U_3^{(N)}) \longrightarrow S_{\text{YM}, \chi}^{\text{cont}}[F^\infty]$$

along every admissible deterministic flat regulator family of that paper.

Proof. The displayed formula is exactly Definition ???. Specializing to the sampled deterministic plaquette holonomies of [?] identifies the right-hand side with the localized Wilson functional $S_{\mathcal{W}, \chi}^{(N)}$. The convergence statement is therefore precisely the local continuum-limit theorem proved in that paper. $\square$

Corollary 1.9 (Regulator-independence of the distinguished Wilson sector in the sampled deterministic presentation). *Within the admissible deterministic flat regulator class of [?], the limit of $j_N(\mathcal{W}_\chi)(U_3^{(N)})$ depends only on χ and the continuum curvature profile F^∞ , not on the chosen admissible regulator exhaustion.*

Proof. Theorem ?? identifies every admissible-sequence limit with the same explicit continuum quantity

$$S_{\text{YM}, \chi}^{\text{cont}}[F^\infty].$$

Hence the limit is independent of the particular admissible deterministic flat regulator family. $\square$

Corollary 1.10 (Sampled deterministic continuum Wilson functional on the distinguished local sector). *The assignment*

$$\omega_{\mathcal{W}}^\infty(\mathcal{W}_\chi) := S_{\text{YM}, \chi}^{\text{cont}}[F^\infty]$$

defines a linear gauge-invariant continuum functional on the linear span of the symbols $\{\mathcal{W}_\chi : \chi \in C_c(\mathbb{R}^4)\}$ associated with the sampled deterministic holonomy family of [?], and the deterministic finite-regulator representatives $j_N(\mathcal{W}_\chi)(U_3^{(N)})$ converge to that functional along every admissible deterministic flat regulator family.

Proof. Linearity in χ is immediate from the definition of $S_{\text{YM}, \chi}^{\text{cont}}[F^\infty]$. The convergence statement is exactly Theorem ???. $\square$

2 Configuration space and measure

2.1 Imported lattice data

We summarize only the regulator-gauge ingredients from [?] and the explicit matter conventions adopted in this paper that are required for quantization.

Definition 2.1 (Imported lattice package). *For each regulator size N , the deterministic regulator construction supplies the finite regulator graph and Wilson sector*

$$(\mathcal{N}, \mathcal{E}, \mathcal{C}_\square) = (\mathcal{N}^{(N)}, \mathcal{E}^{(N)}, \mathcal{C}_\square^{(N)})$$

and we equip that graph with the following finite ingredients:

1. finite site and oriented edge sets $\mathcal{N}, \mathcal{E}$;
2. for every plaquette $P \in \mathcal{C}_\square$, an oriented boundary

$$\partial P = (e_1, e_2, e_3, e_4), \quad e_k \in \mathcal{E},$$

with the ordered plaquette holonomy

$$U(P) := U(e_1)U(e_2)U(e_3)U(e_4);$$

3. the Wilson action

$$S_W(U) := \beta \sum_{P \in \mathcal{C}_\square} \left(1 - \frac{1}{3} \text{Re tr } U(P) \right), \quad \beta := \frac{6}{g_3^2};$$

4. the matter actions

$$S_F(U, \bar{\psi}, \psi) := \sum_{e \in \mathcal{E}} \bar{\psi}_{t(e)} (\psi_{t(e)} - U(e)\psi_{s(e)}) + m_F \sum_{x \in \mathcal{N}} \bar{\psi}_x \psi_x,$$

$$S_\phi(\phi) := \frac{1}{2} \sum_{e=(x \rightarrow y) \in \mathcal{E}} \frac{(\phi_y - \phi_x)^2}{\ell_e + \eta} + \frac{m_\phi^2}{2} \sum_{x \in \mathcal{N}} \phi_x^2,$$

where ℓ_e is the fixed edge length induced by the regulator geometry and $\eta > 0$ is the distance regularization.

Remark 2.2. The gauge sector above is exactly the regulator Wilson sector of [?]. The only change made there is interpretive: the sampled background links are promoted to independent coordinates on $SU(3)^\mathcal{E}$. The fermionic and scalar terms are then fixed explicitly on that same finite graph by the formulas displayed in Definition ??.

2.2 Field manifold

Definition 2.3 (Gauge-link manifold). *Let*

$$\mathcal{U} := SU(3)^\mathcal{E} = \{U : \mathcal{E} \rightarrow SU(3)\}.$$

Because $\mathcal{E}$ is finite and $SU(3)$ is a compact smooth Lie group of real dimension 8, $\mathcal{U}$ is a compact smooth manifold of real dimension $8|\mathcal{E}|$.

Definition 2.4 (Auxiliary color manifold). *Let*

$$\mathbb{B} := \{z \in \mathbb{C}^3 : \|z\| < 1\}, \quad \mathcal{C} := \mathbb{B}^\mathcal{N}.$$

This is the product of the open unit balls suggested by the color normalization bound $\|\mathbf{c}_x\| < 1$ proved in [?]. Since no independent color action was supplied in the companion paper, this sector will be assigned a normalized product measure and decouples from the dynamical integrals below.

Definition 2.5 (Matter field domains). *Let*

$$\mathcal{S} := \mathbb{R}^{\mathcal{N}}$$

be the scalar field space. For the fermionic sector, write

$$V_F := \bigoplus_{x \in \mathcal{N}} \mathbb{C}_x^3$$

and let Λ_F be the Grassmann algebra generated by

$$\{\psi_{x,\alpha}, \bar{\psi}_{x,\alpha} : x \in \mathcal{N}, \alpha = 1, 2, 3\}.$$

The full promoted field manifold is

$$\mathcal{X}_{\mathcal{L}} := \mathcal{U} \times \mathcal{C} \times \Lambda_F \times \mathcal{S}.$$

Remark 2.6. Only the gauge and scalar sectors are manifolds in the ordinary sense. The fermionic sector is finite-dimensional Grassmann algebra, so integration in those variables is algebraic rather than measure theoretic. This is standard for finite lattice fermions [?, ?].

2.3 Measures

Definition 2.7 (Normalized Haar measure). *Let μ_H denote the unique normalized Haar probability measure on $SU(3)$, so that*

$$\mu_H(SU(3)) = 1$$

and for every Borel set $A \subseteq SU(3)$ and every $g \in SU(3)$,

$$\mu_H(gA) = \mu_H(Ag) = \mu_H(A).$$

The gauge-link measure on $\mathcal{U}$ is the product probability measure

$$d\mu_{\mathcal{U}}(U) := \prod_{e \in \mathcal{E}} d\mu_H(U(e)).$$

Definition 2.8 (Auxiliary color measure). *Let $\nu_{\mathbb{B}}$ be normalized Lebesgue measure on $\mathbb{B} \subset \mathbb{C}^3 \cong \mathbb{R}^6$. The color measure on $\mathcal{C}$ is*

$$d\nu_{\mathcal{C}}(c) := \prod_{x \in \mathcal{N}} d\nu_{\mathbb{B}}(c_x).$$

By normalization, $\nu_{\mathcal{C}}(\mathcal{C}) = 1$.

Definition 2.9 (Berezin and scalar measures). *The Berezin measure is*

$$d\mu_F := \prod_{x \in \mathcal{N}} \prod_{\alpha=1}^3 d\bar{\psi}_{x,\alpha} d\psi_{x,\alpha},$$

with the convention

$$\int d\bar{\psi} d\psi e^{-\bar{\psi}\psi} = 1.$$

The scalar measure on $\mathcal{S}$ is ordinary Lebesgue measure

$$d\phi := \prod_{x \in \mathcal{N}} d\phi_x.$$

Definition 2.10 (Total finite-lattice measure). *The total measure symbol on $\mathcal{X}_{\mathcal{L}}$ is*

$$d\mu := d\mu_{\mathcal{U}}(U) d\nu_{\mathcal{C}}(c) d\mu_{\mathbf{F}} d\phi.$$

When the integrand does not depend on c , the auxiliary color sector contributes the unit factor

$$\int_{\mathcal{C}} d\nu_{\mathcal{C}}(c) = 1.$$

Proposition 2.11 (Well-posedness of the finite measure factors). *Each factor in Definition ?? is well defined. In particular, $\mu_{\mathcal{U}}$ is a Borel probability measure on a compact manifold, $\nu_{\mathcal{C}}$ is a Borel probability measure on a bounded open subset of $\mathbb{R}^{6|\mathcal{N}|}$, $\mu_{\mathbf{F}}$ is a finite Berezin integration rule on a finite Grassmann algebra, and $\phi \mapsto S_{\phi}(\phi)$ is a measurable function on $\mathbb{R}^{|\mathcal{N}|}$.*

Proof. The product group $SU(3)^{\mathcal{E}}$ is compact because $\mathcal{E}$ is finite and $SU(3)$ is compact, hence the product Haar measure exists. The set $\mathcal{C} = \mathbb{B}^{\mathcal{N}}$ is a bounded open subset of a finite-dimensional Euclidean space, so the normalized Lebesgue measure exists. The fermionic sector has finitely many generators, hence finite Berezin integration is an algebraic operation. Finally S_{ϕ} is a finite sum of continuous functions of ϕ , because each coefficient $(\ell_e + \eta)^{-1}$ is finite and positive. $\square$

3 Construction of the partition function

The total Euclidean action. The quantized action on $\mathcal{X}_{\mathcal{L}}$ is the same algebraic expression as in the companion lattice paper, but now interpreted on the promoted field space:

$$S(U, \bar{\psi}, \psi, \phi) := S_{\mathbf{W}}(U) + S_{\mathbf{F}}(U, \bar{\psi}, \psi) + S_{\phi}(\phi).$$

The auxiliary color variable $c \in \mathcal{C}$ does not enter this action and is therefore normalized away. No additional asymptotic limit is taken at this stage. Every term is a finite sum over the fixed finite index sets $\mathcal{N}$, $\mathcal{E}$, and $\mathcal{C}_{\square}$.

The master partition function. The central object is the finite-lattice partition function

$$Z := \int_{\mathcal{X}_{\mathcal{L}}} \exp(-S(U, \bar{\psi}, \psi, \phi)) d\mu. \quad (1)$$

Because the color sector is normalized and decoupled,

$$Z = \int_{\mathcal{U} \times \Lambda_{\mathbf{F}} \times \mathcal{S}} \exp(-S_{\mathbf{W}}(U) - S_{\mathbf{F}}(U, \bar{\psi}, \psi) - S_{\phi}(\phi)) d\mu_{\mathcal{U}}(U) d\mu_{\mathbf{F}} d\phi.$$

Unnormalized Euclidean functionals. For any bounded gauge-invariant observable $\mathcal{O}$ whose fermionic degree is even, define the unnormalized Euclidean functional by

$$\tilde{\omega}_N(\mathcal{O}) := \int_{\mathcal{X}_{\mathcal{L}}} \mathcal{O}(U, c, \bar{\psi}, \psi, \phi) e^{-S(U, \bar{\psi}, \psi, \phi)} d\mu. \quad (2)$$

Normalization and expectation values. When $Z \neq 0$, the normalized vacuum expectation is defined by

$$\langle \mathcal{O} \rangle := \frac{1}{Z} \tilde{\omega}_N(\mathcal{O}), \quad (3)$$

provided $Z \neq 0$. The nonvanishing of Z is model dependent and cannot be asserted for arbitrary finite complex fermion determinants. What can always be proved is finiteness of the numerator and denominator as complex numbers, which is sufficient for all formal manipulations before division by Z .

Theorem 3.1 (Existence and finiteness of the partition function). *Under the hypotheses of Theorem ??, the partition function (??) is a finite complex number. Moreover, for every bounded measurable observable $\mathcal{O}$ depending polynomially on the Grassmann variables, the unnormalized expectation*

$$\int_{\mathcal{X}_{\mathcal{L}}} \mathcal{O} e^{-S} d\mu$$

is finite.

Proof. Integrate the fermions first. Section ?? will show that

$$\int_{\Lambda_F} e^{-S_F(U, \bar{\psi}, \psi)} d\mu_F = \det M_F(U),$$

where $M_F(U)$ is a finite complex matrix depending continuously on U . Since $\mathcal{U}$ is compact, the function $U \mapsto \det M_F(U)$ is bounded:

$$\sup_{U \in \mathcal{U}} |\det M_F(U)| < \infty.$$

The Wilson term satisfies $S_W(U) \geq 0$ for every U , because each summand $1 - \frac{1}{3} \operatorname{Re} \operatorname{tr} U(P)$ is nonnegative. Hence

$$|e^{-S_W(U)}| \leq 1.$$

For the scalar sector,

$$S_\phi(\phi) \geq \frac{m_\phi^2}{2} \sum_{x \in \mathcal{N}} \phi_x^2,$$

because the edge term is nonnegative. Therefore

$$e^{-S_\phi(\phi)} \leq \exp\left(-\frac{m_\phi^2}{2} \sum_{x \in \mathcal{N}} \phi_x^2\right),$$

and the right-hand side is integrable on $\mathbb{R}^{|\mathcal{N}|}$. Putting the bounds together,

$$|Z| \leq \sup_{U \in \mathcal{U}} |\det M_F(U)| \int_{\mathbb{R}^{|\mathcal{N}|}} e^{-\frac{m_\phi^2}{2} \sum_x \phi_x^2} d\phi < \infty.$$

The same domination argument works after multiplying by any bounded observable that is polynomial in the finitely many Grassmann generators, because Berezin integration of such a polynomial again yields a bounded continuous function of U . $\square$

4 Gauge redundancy and orbit analysis

The gauge orbit problem. The finite lattice theory has a local gauge symmetry at every site. This means that the coordinates $(U, \psi, \bar{\psi}, c)$ on field space are not physical coordinates: many different tuples encode the same gauge state. The redundancy is continuous, because the gauge group is a product of copies of the compact Lie group $SU(3)$.

Definition 4.1 (Local gauge group). *The lattice gauge group is*

$$\mathcal{G} := SU(3)^{\mathcal{N}} = \{\Omega : \mathcal{N} \rightarrow SU(3)\}.$$

Its action on the promoted fields is

$$U^\Omega(e) := \Omega(t(e)) U(e) \Omega(s(e))^{-1},$$

$$\psi_x^\Omega := \Omega(x) \psi_x, \quad \bar{\psi}_x^\Omega := \bar{\psi}_x \Omega(x)^{-1}, \quad c_x^\Omega := \Omega(x) c_x, \quad \phi_x^\Omega := \phi_x.$$

Proposition 4.2 (Gauge invariance of action and measure). *For every $\Omega \in \mathcal{G}$,*

$$S_W(U^\Omega) = S_W(U), \quad S_F(U^\Omega, \bar{\psi}^\Omega, \psi^\Omega) = S_F(U, \bar{\psi}, \psi), \quad S_\phi(\phi^\Omega) = S_\phi(\phi).$$

Moreover, $\mu_{\mathcal{U}}$, $\nu_{\mathcal{E}}$, and μ_F are invariant under the corresponding gauge transformations. Consequently the total weighted measure $e^{-S} d\mu$ is gauge invariant.

Proof. Gauge invariance of S_W was proved in [?]. For the matter sector, a direct calculation from the defining formulas shows that S_F is invariant under

$$\psi_x \mapsto \Omega(x) \psi_x, \quad \bar{\psi}_x \mapsto \bar{\psi}_x \Omega(x)^{-1},$$

because the gauge factors cancel pointwise in each edge term and in each mass term, while S_ϕ is invariant because the scalar field is gauge neutral. Haar invariance of $\mu_{\mathcal{U}}$ follows edgewise from left and right invariance of Haar measure on $SU(3)$. The normalized Lebesgue measure on the ball $\mathbb{B} \subset \mathbb{C}^3$ is invariant under the unitary action of $SU(3)$ because the action preserves Euclidean norms and Jacobian 1. Berezin measure is invariant under the linear change of variables $(\psi, \bar{\psi}) \mapsto (\Omega\psi, \bar{\psi}\Omega^{-1})$ because the Jacobians $\det \Omega$ and $\det \Omega^{-1}$ cancel and $\det \Omega = 1$ in $SU(3)$. $\square$

Definition 4.3 (Gauge orbit and stabilizer). *For a field configuration*

$$\Phi := (U, c, \bar{\psi}, \psi, \phi) \in \mathcal{X}_{\mathcal{L}}$$

its gauge orbit is

$$\mathcal{O}(\Phi) := \{\Phi^\Omega : \Omega \in \mathcal{G}\},$$

and its stabilizer is

$$\text{Stab}(\Phi) := \{\Omega \in \mathcal{G} : \Phi^\Omega = \Phi\}.$$

Symmetry redundancy versus physical degrees of freedom. The physical configuration space is not $\mathcal{X}_{\mathcal{L}}$ itself but the orbit space $\mathcal{X}_{\mathcal{L}}/\mathcal{G}$. On a finite graph one can count the redundant and nonredundant link degrees explicitly. This count is most transparent after choosing a spanning forest.

Definition 4.4 (Underlying undirected graph and component count). *Let $|\mathcal{L}|$ be the undirected graph obtained from $(\mathcal{N}, \mathcal{E})$ by forgetting orientation and identifying parallel opposite orientations when both are present. Let $b_0(|\mathcal{L}|)$ be the number of connected components of $|\mathcal{L}|$.*

Proposition 4.5 (Dimension count for gauge redundancy). *Assume the underlying undirected graph $|\mathcal{L}|$ has $b_0 := b_0(|\mathcal{L}|)$ connected components. Then:*

1. *the gauge-link manifold has real dimension $8|\mathcal{E}|$;*
2. *the rooted gauge group obtained by fixing one vertex per connected component has real dimension $8(|\mathcal{N}| - b_0)$;*
3. *after maximal-tree gauge fixing, the number of remaining independent gauge link coordinates is*

$$8(|\mathcal{E}| - |\mathcal{N}| + b_0),$$

which is 8 times the cycle rank of the graph.

Proof. Item (1) is immediate from $\dim SU(3) = 8$. For item (2), each unfixed site contributes one copy of $SU(3)$. For item (3), every spanning forest on a graph with $|\mathcal{N}|$ vertices and b_0 connected components has $|\mathcal{N}| - b_0$ edges, so removing exactly those tree-edge coordinates leaves $|\mathcal{E}| - |\mathcal{N}| + b_0$ cotree edges. Each cotree edge carries 8 real parameters. $\square$

Remark 4.6. The compactness of $SU(3)$ means that the gauge-group probability measure is finite when normalized. Thus on a finite lattice the problem with gauge redundancy is not an *infinite numerical volume divergence*; the problem is that the parametrization is many-to-one and therefore obscures the physical degrees of freedom. Gauge fixing removes this redundancy exactly.

5 Gauge fixing and the Faddeev–Popov procedure

Selecting the gauge condition. For a finite graph the cleanest globally valid gauge choice is a rooted maximal-tree gauge. Unlike lattice Landau gauge, it has no Gribov ambiguity: every rooted orbit intersects the gauge slice exactly once.

Definition 5.1 (Rooted spanning forest). *Choose one root vertex in each connected component of $|\mathcal{L}|$ and let $R \subseteq \mathcal{N}$ be the set of roots. A rooted spanning forest $\mathcal{T} \subseteq \mathcal{E}$ is an oriented set of edges such that:*

1. *after forgetting orientation, $\mathcal{T}$ is a spanning forest of $|\mathcal{L}|$;*
2. *every edge of $\mathcal{T}$ is oriented away from its component root;*
3. *every nonroot vertex $x \in \mathcal{N} \setminus R$ has a unique parent edge*

$$e_x = (p(x) \rightarrow x) \in \mathcal{T}.$$

Definition 5.2 (Rooted gauge group and maximal-tree gauge slice). *The rooted gauge group is*

$$\mathcal{G}_* := \{\Omega \in \mathcal{G} : \Omega(r) = I \text{ for every root } r \in R\}.$$

The maximal-tree gauge slice is

$$\Sigma_{\mathcal{T}} := \{U \in \mathcal{U} : U(e) = I \text{ for every } e \in \mathcal{T}\}.$$

Theorem 5.3 (Global rooted maximal-tree gauge). *For every $U \in \mathcal{U}$ there exists a unique $\Omega_U \in \mathcal{G}_*$ such that*

$$U^{\Omega_U} \in \Sigma_{\mathcal{T}}.$$

Equivalently, every rooted gauge orbit in $\mathcal{U}$ intersects $\Sigma_{\mathcal{T}}$ exactly once.

Proof. Existence is recursive along each tree component. Set $\Omega_U(r) = I$ at every root. Suppose $\Omega_U(p(x))$ is already defined for the parent of a nonroot vertex x . The condition

$$U^{\Omega_U}(e_x) = I$$

for the parent edge $e_x = (p(x) \rightarrow x)$ is

$$\Omega_U(x) U(e_x) \Omega_U(p(x))^{-1} = I,$$

hence

$$\Omega_U(x) = \Omega_U(p(x)) U(e_x)^{-1}.$$

Because every nonroot vertex has a unique parent and every component contains a unique path from the root to any vertex, this recursion defines Ω_U uniquely on all vertices. By construction $\Omega_U \in \mathcal{G}_*$ and $U^{\Omega_U}(e) = I$ for all $e \in \mathcal{T}$, so $U^{\Omega_U} \in \Sigma_{\mathcal{T}}$.

For uniqueness, let $\Omega_1, \Omega_2 \in \mathcal{G}_*$ satisfy $U^{\Omega_1}, U^{\Omega_2} \in \Sigma_{\mathcal{T}}$. Set $\Gamma := \Omega_2^{-1} \Omega_1 \in \mathcal{G}_*$. Then for every tree edge $e = (x \rightarrow y) \in \mathcal{T}$,

$$I = (U^{\Omega_1})(e) = \Gamma(y) (U^{\Omega_2})(e) \Gamma(x)^{-1} = \Gamma(y) \Gamma(x)^{-1}.$$

Thus $\Gamma(y) = \Gamma(x)$ on every tree edge. Since $\Gamma(r) = I$ on every root and every vertex is connected to a root by a unique tree path, induction along the tree gives $\Gamma(x) = I$ for all x . Hence $\Omega_1 = \Omega_2$. $\square$

The Faddeev–Popov identity. Because the gauge slice is global and unique, the Faddeev–Popov formula is particularly simple.

Definition 5.4 (Delta distribution on the tree gauge slice). *Let δ_I be the Dirac distribution at the identity of $SU(3)$, defined by*

$$\int_{SU(3)} f(V) \delta_I(V) d\mu_H(V) = f(I)$$

for continuous f . Define the tree-gauge delta distribution by

$$\delta_{\mathcal{T}}(U) := \prod_{e \in \mathcal{T}} \delta_I(U(e)).$$

Theorem 5.5 (Exact Faddeev–Popov identity in maximal-tree gauge). *For every $U \in \mathcal{U}$,*

$$1 = \int_{\mathcal{G}_*} \delta_{\mathcal{T}}(U^{\Omega}) d\mu_{\mathcal{G}_*}(\Omega),$$

where $\mu_{\mathcal{G}_*}$ is product normalized Haar measure on $\mathcal{G}_*$. Hence for every gauge-invariant integrable functional F on $\mathcal{U}$,

$$\int_{\mathcal{U}} F(U) d\mu_{\mathcal{U}}(U) = \int_{\mathcal{U}} F(U) \delta_{\mathcal{T}}(U) d\mu_{\mathcal{U}}(U).$$

Proof. Fix $U \in \mathcal{U}$. We prove the first identity by induction on the number of nonroot vertices. If the rooted forest has no nonroot vertices, then $\mathcal{T} = \emptyset$ and $\delta_{\mathcal{T}} \equiv 1$, so the claim is immediate. Assume now that $\mathcal{T}$ has at least one edge and choose a leaf $x \in \mathcal{N} \setminus R$. Because x has no children, the only factor in $\delta_{\mathcal{T}}(U^{\Omega})$ that depends on $\Omega(x)$ is the factor associated with the parent edge $e_x = (p(x) \rightarrow x)$:

$$\delta_I(U^{\Omega}(e_x)) = \delta_I(\Omega(x) U(e_x) \Omega(p(x))^{-1}).$$

For fixed values of all other gauge variables, set $g_x := U(e_x)\Omega(p(x))^{-1} \in SU(3)$. Then left invariance of Haar measure gives

$$\int_{SU(3)} \delta_I(\Omega(x)g_x) \, d\mu_H(\Omega(x)) = \int_{SU(3)} \delta_I(V) \, d\mu_H(V) = 1.$$

Thus integrating over the leaf variable contributes exactly the unit factor and removes the leaf edge from the product. Deleting the leaf and repeating the same argument inductively over the smaller rooted forest proves

$$\int_{\mathcal{G}_*} \delta_{\mathcal{T}}(U^\Omega) \, d\mu_{\mathcal{G}_*}(\Omega) = 1.$$

For the second identity, insert the first into the integral of F :

$$\int_{\mathcal{U}} F(U) \, d\mu_{\mathcal{U}}(U) = \int_{\mathcal{U}} \int_{\mathcal{G}_*} F(U) \delta_{\mathcal{T}}(U^\Omega) \, d\mu_{\mathcal{G}_*}(\Omega) \, d\mu_{\mathcal{U}}(U).$$

Use Fubini's theorem and then change variables $U \mapsto U^{\Omega^{-1}}$. Haar invariance implies $d\mu_{\mathcal{U}}(U^{\Omega^{-1}}) = d\mu_{\mathcal{U}}(U)$, and gauge invariance of F gives $F(U^{\Omega^{-1}}) = F(U)$. Thus

$$\int_{\mathcal{U}} F(U) \, d\mu_{\mathcal{U}}(U) = \int_{\mathcal{U}} F(U) \delta_{\mathcal{T}}(U) \, d\mu_{\mathcal{U}}(U) \int_{\mathcal{G}_*} d\mu_{\mathcal{G}_*}(\Omega).$$

The last factor equals 1 because $\mu_{\mathcal{G}_*}$ is normalized. □

The Faddeev–Popov determinant. The determinant can be computed directly from infinitesimal gauge variations. In maximal-tree gauge it is constant.

Definition 5.6 (Infinitesimal gauge parameters). *Let*

$$T^a := \frac{\lambda_a}{2}, \quad a = 1, \dots, 8,$$

where λ_a are the Gell-Mann matrices. For each site $x \in \mathcal{N}$ let

$$\omega_x = \sum_{a=1}^8 \omega_x^a T^a \in \mathfrak{h}_3,$$

and define the infinitesimal rooted gauge transformation

$$\Omega_\varepsilon(x) := \exp(i\varepsilon \omega_x), \quad \omega_r = 0 \text{ for every } r \in R.$$

Proposition 5.7 (Linearized gauge map on the tree). *Let $U \in \Sigma_{\mathcal{T}}$, so $U(e) = I$ for all $e \in \mathcal{T}$. For a tree edge $e = (x \rightarrow y) \in \mathcal{T}$,*

$$-i \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} U^{\Omega_\varepsilon}(e) = \omega_y - \omega_x.$$

In the basis $\{T^a\}_{a=1}^8$, the linearized Faddeev–Popov operator is

$$M_{\text{FP}}^{(\mathcal{T})} = B_{\mathcal{T}} \otimes I_8,$$

where $B_{\mathcal{T}}$ is the rooted incidence matrix of the spanning forest.

Proof. Differentiate

$$U^{\Omega_\varepsilon}(e) = \Omega_\varepsilon(y) U(e) \Omega_\varepsilon(x)^{-1}$$

at $\varepsilon = 0$. Since $U(e) = I$ on the slice,

$$-i \frac{d}{d\varepsilon} \Big|_{\varepsilon=0} U^{\Omega_\varepsilon}(e) = \omega_y - \omega_x.$$

Writing each ω_x in the Gell-Mann basis shows that the eight color components decouple and each is acted on by the same incidence matrix $B_{\mathcal{T}}$. $\square$

Proposition 5.8 (The Faddeev–Popov determinant is identically one). *With the rooted maximal-tree gauge choice,*

$$\Delta_{\text{FP}}^{(\mathcal{T})} = \det M_{\text{FP}}^{(\mathcal{T})} = 1.$$

Proof. Order the nonroot vertices so that every parent precedes every child. Index the rows of $B_{\mathcal{T}}$ by the tree edges $e_x = (p(x) \rightarrow x)$ and the columns by the nonroot vertices x . Then the row corresponding to e_x has entry $+1$ in column x , entry -1 in column $p(x)$ if $p(x)$ is nonroot, and zero in all columns associated with vertices not on the path from the root to x . By the chosen ordering this matrix is triangular with diagonal entries all equal to 1. Hence

$$\det B_{\mathcal{T}} = 1.$$

Therefore

$$\det M_{\text{FP}}^{(\mathcal{T})} = \det(B_{\mathcal{T}} \otimes I_8) = (\det B_{\mathcal{T}})^8 = 1.$$

$\square$

Corollary 5.9 (Gauge-fixed partition function). *The partition function may be written as*

$$Z = \int_{\mathcal{U} \times \Lambda_{\text{F}} \times \mathcal{S}} \delta_{\mathcal{T}}(U) e^{-S_{\text{W}}(U) - S_{\text{F}}(U, \bar{\psi}, \psi) - S_{\phi}(\phi)} d\mu_{\mathcal{U}}(U) d\mu_{\text{F}} d\phi.$$

After integrating out the constrained tree links, this becomes an ordinary integral over the cotree edge variables

$$\mathcal{E}_{\text{co}} := \mathcal{E} \setminus \mathcal{T}.$$

Proof. Apply Theorem ?? to the gauge-invariant integrand obtained after combining the matter variables with Proposition ?. Since $\delta_{\mathcal{T}}(U)$ forces $U(e) = I$ on all tree edges, integrating over those edges contributes the unit factor

$$\int_{SU(3)} \delta_I(V) d\mu_H(V) = 1$$

edge by edge. $\square$

6 Ghost fields and the effective action

Introducing ghost fields. Although the Faddeev–Popov determinant is trivial in rooted maximal-tree gauge, it is still useful to represent it as a finite Berezin integral, because this makes the gauge-fixing algebra parallel to the standard continuum formalism. The continuum-indexed gauge-invariant observable package fixed in Definitions ?? and ?? is independent of these auxiliary variables; the present section records the exact finite-dimensional algebraic repackaging of the same gauge-fixed measure.

Definition 6.1 (Ghost and antighost fields). *For each nonroot vertex $x \in \mathcal{N} \setminus R$, let*

$$c_x = \sum_{a=1}^8 c_x^a T^a$$

be an anticommuting $\mathfrak{h}_3$ -valued ghost field. For each tree edge $e \in \mathcal{T}$, let

$$\bar{c}_e = \sum_{a=1}^8 \bar{c}_e^a T^a$$

be an anticommuting $\mathfrak{h}_3$ -valued antighost field. Their Berezin measure is

$$d\mu_{\text{gh}} := \prod_{e \in \mathcal{T}} \prod_{a=1}^8 d\bar{c}_e^a \prod_{x \in \mathcal{N} \setminus R} \prod_{a=1}^8 dc_x^a.$$

For every root vertex $r \in R$, set

$$c_r := 0.$$

Definition 6.2 (Ghost action in maximal-tree gauge). *Define*

$$S_{\text{ghost}}^{(\mathcal{T})}(\bar{c}, c) := 2 \sum_{e=(x \rightarrow y) \in \mathcal{T}} \text{tr}(\bar{c}_e(c_y - c_x)),$$

with the convention $c_r := 0$ for root vertices $r \in R$.

Proposition 6.3 (Ghost representation of the Faddeev–Popov determinant). *The ghost integral reproduces the Faddeev–Popov determinant:*

$$\int e^{-S_{\text{ghost}}^{(\mathcal{T})}(\bar{c}, c)} d\mu_{\text{gh}} = \Delta_{\text{FP}}^{(\mathcal{T})} = 1.$$

Proof. Expand the ghost fields in the basis $\{T^a\}_{a=1}^8$. Then the action factorizes over color indices:

$$S_{\text{ghost}}^{(\mathcal{T})}(\bar{c}, c) = \sum_{a=1}^8 \sum_{e=(x \rightarrow y) \in \mathcal{T}} \bar{c}_e^a (c_y^a - c_x^a).$$

Here we used $\text{tr}(T^a T^b) = \frac{1}{2} \delta_{ab}$. This is precisely the quadratic Berezin form associated with the matrix $B_{\mathcal{T}} \otimes I_8$, so the standard finite-dimensional Berezin identity gives

$$\int e^{-S_{\text{ghost}}^{(\mathcal{T})}} d\mu_{\text{gh}} = \det(B_{\mathcal{T}} \otimes I_8).$$

Proposition ?? gives the value 1. □

The effective quantized action. To display the standard BRST structure it is convenient to introduce auxiliary bosonic Nakanishi–Lautrup fields $b_e \in \mathfrak{h}_3$ for $e \in \mathcal{T}$.

Definition 6.4 (BRST differential and gauge-fixing fermion). *Let s act on the full field algebra by*

$$\begin{aligned} s U(e) &= i c_{t(e)} U(e) - i U(e) c_{s(e)}, & s \psi_x &= i c_x \psi_x, & s \bar{\psi}_x &= -i \bar{\psi}_x c_x, \\ s \phi_x &= 0, & s c_x &= -i c_x^2, & s \bar{c}_e &= b_e, & s b_e &= 0. \end{aligned}$$

Here the rooted ghost convention from Definition ?? is in force, so $c_r = 0$ for every root $r \in R$. Define the gauge-fixing fermion

$$\Psi_{\mathcal{T}} := \sum_{e \in \mathcal{T}} 2 \text{tr} \left(\bar{c}_e \left(-i(U(e) - I) + \frac{\alpha}{2} b_e \right) \right), \quad \alpha \geq 0.$$

Lemma 6.5 (Nilpotency of the BRST differential). *The operator s satisfies $s^2 = 0$ on generators and therefore on the whole field algebra.*

Proof. For ϕ_x and b_e the claim is immediate. For $\bar{c}_e$, $s^2\bar{c}_e = sb_e = 0$. For c_x ,

$$s^2c_x = -i s(c_x^2) = -i((sc_x)c_x - c_x(sc_x)) = -i((-ic_x^2)c_x - c_x(-ic_x^2)) = 0.$$

For ψ_x ,

$$s^2\psi_x = i(sc_x)\psi_x - ic_x(s\psi_x) = i(-ic_x^2)\psi_x - ic_x(ic_x\psi_x) = 0,$$

and similarly for $\bar{\psi}_x$. Finally, for a link variable $U(e)$,

$$\begin{aligned} s^2U(e) &= i(sc_{t(e)})U(e) + ic_{t(e)}sU(e) - isU(e)c_{s(e)} - iU(e)sc_{s(e)} \\ &= c_{t(e)}^2U(e) - c_{t(e)}^2U(e) + U(e)c_{s(e)}^2 - U(e)c_{s(e)}^2 = 0. \end{aligned}$$

□

Proposition 6.6 (BRST-exact gauge fixing). *The gauge-fixing and ghost terms are*

$$S_{\text{gf+gh}}^{(\mathcal{T})} := s\Psi_{\mathcal{T}}.$$

Explicitly,

$$S_{\text{gf+gh}}^{(\mathcal{T})} = \sum_{e \in \mathcal{T}} 2 \operatorname{tr} \left(-i b_e (U(e) - I) + \frac{\alpha}{2} b_e^2 + \bar{c}_e (c_{t(e)} U(e) - U(e) c_{s(e)}) \right).$$

Consequently the effective action

$$S_{\text{eff}} := S_W + S_F + S_\phi + S_{\text{gf+gh}}^{(\mathcal{T})}$$

is BRST invariant:

$$sS_{\text{eff}} = 0.$$

Proof. Apply s to $\Psi_{\mathcal{T}}$ and use Definition ???. Since the fields b_e commute with everything and the ghosts anticommute, one obtains the displayed formula. Gauge invariance of $S_W + S_F + S_\phi$ implies $s(S_W + S_F + S_\phi) = 0$, because s is the infinitesimal gauge differential with ghost parameter c . Then

$$sS_{\text{eff}} = s(S_W + S_F + S_\phi) + s^2\Psi_{\mathcal{T}} = 0$$

by Lemma ???. □

Remark 6.7. On the maximal-tree gauge slice $U(e) = I$ for $e \in \mathcal{T}$, the ghost term in $S_{\text{gf+gh}}^{(\mathcal{T})}$ reduces exactly to $S_{\text{ghost}}^{(\mathcal{T})}$ from Definition ???. Since the Faddeev–Popov determinant is constant, the ghost sector decouples after gauge fixing. The BRST presentation is therefore an algebraic repackaging of the same finite gauge-fixing data.

7 Matter sector integration

Fermionic Berezin integration. On a finite lattice the fermion integral is exact and elementary once the quadratic form is written in matrix notation.

Definition 7.1 (Fermion matrix). *Order the site-color basis vectors by pairs (x, α) with $x \in \mathcal{N}$ and $\alpha \in \{1, 2, 3\}$. Define the matrix $M_F(U) \in \mathbb{C}^{3|\mathcal{N}| \times 3|\mathcal{N}|}$ by*

$$(M_F(U))_{(y,\alpha),(x,\beta)} := (m_F + \text{indeg}(y)) \delta_{xy} \delta_{\alpha\beta} - \sum_{\substack{e \in \mathcal{E} \\ s(e)=x, t(e)=y}} U(e)_{\alpha\beta},$$

where

$$\text{indeg}(y) := \#\{e \in \mathcal{E} : t(e) = y\}.$$

Lemma 7.2 (Quadratic form identity). *With $M_F(U)$ from Definition ??,*

$$S_F(U, \bar{\psi}, \psi) = \sum_{x,y \in \mathcal{N}} \sum_{\alpha,\beta=1}^3 \bar{\psi}_{y,\alpha} (M_F(U))_{(y,\alpha),(x,\beta)} \psi_{x,\beta}.$$

Proof. Expand the edge sum in Definition ??:

$$\sum_{e \in \mathcal{E}} \bar{\psi}_{t(e)} \psi_{t(e)} = \sum_{y \in \mathcal{N}} \text{indeg}(y) \bar{\psi}_y \psi_y.$$

The remaining transported terms contribute the off-diagonal blocks $-\sum_{e:s(e)=x,t(e)=y} U(e)$. Adding the mass term gives the stated matrix. $\square$

Theorem 7.3 (Fermion determinant). *For every gauge field $U \in \mathcal{U}$,*

$$\int_{\Lambda_F} e^{-S_F(U, \bar{\psi}, \psi)} d\mu_F = \det M_F(U).$$

Consequently

$$Z = \int_{\mathcal{U} \times \mathcal{S}} e^{-S_W(U) - S_\phi(\phi)} \det M_F(U) d\mu_{\mathcal{U}}(U) d\phi.$$

Proof. By Lemma ??, the exponent is the standard quadratic Berezin form $\bar{\Psi} M_F(U) \Psi$ in finitely many Grassmann variables. The finite-dimensional Berezin integral identity therefore gives

$$\int e^{-\bar{\Psi} M \Psi} d\bar{\Psi} d\Psi = \det M$$

for any finite matrix M . Applying this with $M = M_F(U)$ yields the first claim. Substituting the result into the definition of Z gives the second claim. $\square$

Remark 7.4 (Virtual loops and graph support). The determinant $\det M_F(U)$ is a finite polynomial in the link matrices and encodes the usual closed virtual-fermion loops. Because the link set $\mathcal{E}$ is the finite edge set from the deterministic regulator package [?], only loops on this explicit graph contribute. The path support is therefore inherited entirely from that regulator incidence structure, with no additional construction-specific combinatorics appearing in this finite-dimensional determinant expansion.

Scalar kinetic fluctuations. The scalar action is purely bosonic, real valued, and free massive. The regularization $\eta > 0$ excludes edge-length singularities in the kinetic coefficients.

Definition 7.5 (Scalar stiffness operator). *Define the symmetric quadratic form*

$$Q_\eta(\phi) := \frac{1}{2} \sum_{e=(x \rightarrow y) \in \mathcal{E}} \frac{(\phi_y - \phi_x)^2}{\ell_e + \eta}.$$

There is a unique symmetric matrix $L_\eta \in \mathbb{R}^{|\mathcal{N}| \times |\mathcal{N}|}$ such that

$$Q_\eta(\phi) = \frac{1}{2} \phi^\top L_\eta \phi.$$

Lemma 7.6 (Positivity and bounded coefficients). *The operator L_η is positive semidefinite. Moreover every edge weight*

$$w_e := \frac{1}{\ell_e + \eta}$$

satisfies

$$0 < w_e \leq \eta^{-1}.$$

Proof. Each summand in $Q_\eta(\phi)$ is nonnegative, so $Q_\eta(\phi) \geq 0$ for every ϕ , hence L_η is positive semidefinite. Since $\ell_e \geq 0$ and $\eta > 0$,

$$\ell_e + \eta \geq \eta,$$

which implies the upper bound on w_e . □

Theorem 7.7 (Scalar integrability).

$$Z_\phi := \int_{\mathcal{S}} e^{-S_\phi(\phi)} d\phi$$

is finite, and

$$Z_\phi = (2\pi)^{|\mathcal{N}|/2} \det(L_\eta + m_\phi^2 I)^{-1/2}.$$

Proof. By Definition ??,

$$S_\phi(\phi) = \frac{1}{2} \phi^\top L_\eta \phi + \frac{m_\phi^2}{2} \phi^\top \phi.$$

Since L_η is positive semidefinite and $m_\phi^2 I$ is strictly positive, the matrix $L_\eta + m_\phi^2 I$ is positive definite. The standard Gaussian integral formula gives the displayed determinant expression. □

Proposition 7.8 (Scalar source integral). *For every source vector $J = (J_x)_{x \in \mathcal{N}} \in \mathbb{R}^{|\mathcal{N}|}$,*

$$Z_\phi[J] := \int_{\mathcal{S}} \exp\left(-S_\phi(\phi) + \sum_{x \in \mathcal{N}} J_x \phi_x\right) d\phi$$

is finite and satisfies

$$Z_\phi[J] = Z_\phi[0] \exp\left(\frac{1}{2} J^\top (L_\eta + m_\phi^2 I)^{-1} J\right).$$

In particular, all scalar polynomial moments are finite.

Proof. Write

$$K_\phi := L_\eta + m_\phi^2 I.$$

By Theorem ??, the matrix K_ϕ is positive definite. Completing the square gives

$$\frac{1}{2} \phi^\top K_\phi \phi - J^\top \phi = \frac{1}{2} (\phi - K_\phi^{-1} J)^\top K_\phi (\phi - K_\phi^{-1} J) - \frac{1}{2} J^\top K_\phi^{-1} J.$$

Substituting this identity into the integral and translating the integration variable by $K_\phi^{-1} J$ yields the stated formula. Differentiating the right-hand side with respect to the source components proves finiteness of all scalar polynomial moments. $\square$

Remark 7.9 (Role of η). The parameter $\eta > 0$ is not a perturbative convenience; it is the exact mechanism that prevents the scalar kinetic coefficients from diverging when two sites become spacetime-coincident. Without η , the coefficient $1/\ell_e$ would be singular on zero-length edges. With $\eta > 0$, every kinetic coefficient is uniformly bounded by η^{-1} .

8 Lie-algebra coordinates and local measure structure

Local exponential coordinates. The finite-lattice path integral is defined globally by Haar measure on each copy of $SU(3)$. To compare that measure with the usual Lie-algebra variables used in perturbation theory and in local small-field expansions, one works in an exponential chart around the identity.

Definition 8.1 (Symmetric exponential chart). *Let $\mathfrak{h}_3$ be the Hermitian traceless 3×3 matrices with inner product*

$$\langle H, K \rangle := \frac{1}{2} \text{tr}(HK).$$

Choose an open neighborhood $\mathcal{V} \subset \mathfrak{h}_3$ of 0 such that:

1. $\mathcal{V}$ is symmetric: $H \in \mathcal{V}$ implies $-H \in \mathcal{V}$;
2. the map

$$\exp_i : \mathcal{V} \rightarrow SU(3), \quad \exp_i(H) := e^{iH},$$

is a diffeomorphism onto an open neighborhood $\mathcal{U}$ of the identity.

Proposition 8.2 (Single-edge Haar density in exponential coordinates). *Let μ_H be normalized Haar measure on $SU(3)$ and let $\mathcal{V}, \mathcal{U}$ be as in Definition ?? . Then there exists a smooth strictly positive function*

$$J : \mathcal{V} \rightarrow (0, \infty)$$

such that for every continuous function f supported in $\mathcal{U}$,

$$\int_{SU(3)} f(U) \, d\mu_H(U) = \int_{\mathcal{V}} f(e^{iH}) J(H) \, dH,$$

where dH is Lebesgue measure on the 8-dimensional Euclidean space $\mathfrak{h}_3$. After fixing the normalization of dH so that $J(0) = 1$, the density is even and satisfies

$$J(H) = 1 + O(\|H\|^2) \quad \text{as } H \rightarrow 0.$$

Proof. Push forward $\mu_H|_{\mathcal{U}}$ by the inverse chart $\log_i := \exp_i^{-1} : \mathcal{U} \rightarrow \mathcal{V}$. Because $\exp_i$ is a diffeomorphism, the pushforward is a smooth measure on the Euclidean open set $\mathcal{V}$, so it has a smooth strictly positive density $J(H)$ with respect to Lebesgue measure. Since Lebesgue measure on $\mathfrak{h}_3$ is unique only up to a positive constant, we may normalize it so that $J(0) = 1$.

The inversion map on $SU(3)$ sends e^{iH} to e^{-iH} , and Haar measure is invariant under inversion. Because $\mathcal{V}$ is symmetric, this implies

$$J(H) = J(-H) \quad \text{for all } H \in \mathcal{V}.$$

Hence the first-order term in the Taylor expansion of J at the origin vanishes, and smoothness gives

$$J(H) = J(0) + O(\|H\|^2) = 1 + O(\|H\|^2).$$

□

Corollary 8.3 (Second-order matching with Lebesgue measure). *In the chart of Definition ??,*

$$d\mu_H(e^{iH}) = (1 + O(\|H\|^2)) dH \quad \text{as } H \rightarrow 0.$$

Equivalently, for every bounded set $B \subset \mathfrak{h}_3$ and every $a > 0$ small enough,

$$d\mu_H(e^{iaA}) = a^8(1 + O(a^2)) dA$$

uniformly for $A \in B$.

Proof. The first statement is a restatement of Proposition ??. For the second, substitute $H = aA$. Since $\mathfrak{h}_3$ has real dimension 8,

$$dH = a^8 dA,$$

and for A in a fixed bounded set the estimate

$$J(aA) = 1 + O(a^2\|A\|^2) = 1 + O(a^2)$$

is uniform. □

Remark 8.4 (Thermodynamic role of the local chart). The local coordinate description of Haar measure is a fixed- N statement on a single edge. In the broader thermodynamic programme it supplies the local measure model for the finite-lattice link and plaquette observables that appear in the $N \rightarrow \infty$ limit described in [?, ?].

Proposition 8.5 (Transfer of the mean-field LSI to thermodynamic observables). *Let π_∞ be the stationary mean-field law from [?]. Assume, as proved there, that there exist constants $a, b, c \in \mathbb{R}$ with $ac > b^2$ and a constant $C_{\text{LSI}} < \infty$ such that*

$$\text{Ent}_{\pi_\infty}(g^2) \leq C_{\text{LSI}} \int_{\mathbb{R}^8} \Gamma_{\text{hypo}}^\infty(g, g) d\pi_\infty$$

for every smooth compactly supported $g : \mathbb{R}^8 \rightarrow \mathbb{R}$, where

$$\Gamma_{\text{hypo}}^\infty(g, g) := a\|\nabla_v g\|^2 + 2b \nabla_x g \cdot \nabla_v g + c\|\nabla_x g\|^2.$$

For each $k \geq 1$, let

$$\Pi_k^\infty := \pi_\infty^{\otimes k},$$

and define

$$\Gamma_{\text{hypo}}^{(k)}(F, F) := \sum_{j=1}^k \Gamma_{\text{hypo},j}^{\infty}(F, F),$$

where $\Gamma_{\text{hypo},j}^{\infty}$ acts on the j -th variable $z_j = (x_j, v_j) \in \mathbb{R}^8$. Then

$$\text{Ent}_{\Pi_k^{\infty}}(F^2) \leq C_{\text{LSI}} \int_{(\mathbb{R}^8)^k} \Gamma_{\text{hypo}}^{(k)}(F, F) \, d\Pi_k^{\infty}$$

for every smooth compactly supported $F : (\mathbb{R}^8)^k \rightarrow \mathbb{R}$. In particular, the finite-lattice link observables and plaquette observables of [?], which are evaluated on $\pi_{\infty}^{\otimes 2}$ and $\pi_{\infty}^{\otimes 4}$, are supported on equilibrium product measures carrying the same logarithmic Sobolev constant C_{LSI} .

Proof. We argue by induction on k . The case $k = 1$ is exactly the mean-field logarithmic Sobolev inequality proved in [?].

Assume the claim has been proved for $k - 1$, and write

$$z = (z', z_k) \in (\mathbb{R}^8)^{k-1} \times \mathbb{R}^8.$$

For a smooth compactly supported $F : (\mathbb{R}^8)^k \rightarrow \mathbb{R}$, set

$$F_{z_k}(z') := F(z', z_k), \quad m(z_k) := \int_{(\mathbb{R}^8)^{k-1}} F(z', z_k)^2 \, d\Pi_{k-1}^{\infty}(z'), \quad G(z_k) := m(z_k)^{1/2}.$$

The entropy chain rule for product measures gives

$$\text{Ent}_{\Pi_k^{\infty}}(F^2) = \int_{\mathbb{R}^8} \text{Ent}_{\Pi_{k-1}^{\infty}}(F_{z_k}^2) \, d\pi_{\infty}(z_k) + \text{Ent}_{\pi_{\infty}}(G^2).$$

Applying the induction hypothesis to the first term yields

$$\int_{\mathbb{R}^8} \text{Ent}_{\Pi_{k-1}^{\infty}}(F_{z_k}^2) \, d\pi_{\infty}(z_k) \leq C_{\text{LSI}} \sum_{j=1}^{k-1} \int_{(\mathbb{R}^8)^k} \Gamma_{\text{hypo},j}^{\infty}(F, F) \, d\Pi_k^{\infty}.$$

For the second term, regularize the square root by

$$G_{\varepsilon}(z_k) := (m(z_k) + \varepsilon)^{1/2}, \quad \varepsilon > 0.$$

Because F is smooth and compactly supported, the function m is smooth and bounded, hence G_{ε} is smooth and bounded on $\mathbb{R}^8$. Let

$$M := \begin{pmatrix} cI_4 & bI_4 \\ bI_4 & aI_4 \end{pmatrix},$$

so that

$$\Gamma_{\text{hypo},k}^{\infty}(F, F) = \langle \nabla_{z_k} F, M \nabla_{z_k} F \rangle, \quad \nabla_{z_k} := (\nabla_{x_k}, \nabla_{v_k}).$$

Because $ac > b^2$, the matrix M is positive definite. On the set where $m(z_k) > 0$,

$$\nabla_{z_k} G_{\varepsilon}(z_k) = \frac{1}{G_{\varepsilon}(z_k)} \int_{(\mathbb{R}^8)^{k-1}} F(z', z_k) \nabla_{z_k} F(z', z_k) \, d\Pi_{k-1}^{\infty}(z').$$

Therefore

$$G_{\varepsilon}(z_k)^2 \Gamma_{\text{hypo}}^{\infty}(G_{\varepsilon}, G_{\varepsilon}) = \left\langle \int F \nabla_{z_k} F \, d\Pi_{k-1}^{\infty}, M \int F \nabla_{z_k} F \, d\Pi_{k-1}^{\infty} \right\rangle$$

$$\begin{aligned}
&\leq \left(\int F^2 \, d\Pi_{k-1}^\infty \right) \left(\int \langle \nabla_{z_k} F, M \nabla_{z_k} F \rangle \, d\Pi_{k-1}^\infty \right) \\
&= m(z_k) \int_{(\mathbb{R}^8)^{k-1}} \Gamma_{\text{hypo},k}^\infty(F, F)(z', z_k) \, d\Pi_{k-1}^\infty(z').
\end{aligned}$$

Hence

$$\begin{aligned}
\Gamma_{\text{hypo}}^\infty(G_\varepsilon, G_\varepsilon)(z_k) &\leq \frac{m(z_k)}{m(z_k) + \varepsilon} \int_{(\mathbb{R}^8)^{k-1}} \Gamma_{\text{hypo},k}^\infty(F, F)(z', z_k) \, d\Pi_{k-1}^\infty(z') \leq \\
&\int_{(\mathbb{R}^8)^{k-1}} \Gamma_{\text{hypo},k}^\infty(F, F)(z', z_k) \, d\Pi_{k-1}^\infty(z').
\end{aligned}$$

To invoke the one-particle inequality from [?], choose $\chi_R \in C_c^\infty(\mathbb{R}^8)$ such that

$$0 \leq \chi_R \leq 1, \quad \chi_R \equiv 1 \text{ on } B_R, \quad \text{supp } \chi_R \subset B_{2R}, \quad \|\nabla \chi_R\|_\infty \leq \frac{C_\chi}{R}.$$

Set

$$H_{\varepsilon,R} := \chi_R G_\varepsilon \in C_c^\infty(\mathbb{R}^8).$$

Applying the one-particle inequality to $H_{\varepsilon,R}$ gives

$$\text{Ent}_{\pi_\infty}(H_{\varepsilon,R}^2) \leq C_{\text{LSI}} \int_{\mathbb{R}^8} \Gamma_{\text{hypo}}^\infty(H_{\varepsilon,R}, H_{\varepsilon,R}) \, d\pi_\infty.$$

Since G_ε is bounded and $\chi_R \rightarrow 1$ pointwise, dominated convergence gives

$$\text{Ent}_{\pi_\infty}(H_{\varepsilon,R}^2) \rightarrow \text{Ent}_{\pi_\infty}(G_\varepsilon^2) \quad \text{as } R \rightarrow \infty.$$

Moreover,

$$\nabla H_{\varepsilon,R} = \chi_R \nabla G_\varepsilon + G_\varepsilon \nabla \chi_R,$$

so positivity of M implies

$$\Gamma_{\text{hypo}}^\infty(H_{\varepsilon,R}, H_{\varepsilon,R}) \leq 2\chi_R^2 \Gamma_{\text{hypo}}^\infty(G_\varepsilon, G_\varepsilon) + 2G_\varepsilon^2 \Gamma_{\text{hypo}}^\infty(\chi_R, \chi_R).$$

Because a, b, c are fixed, there exists a constant $C_M < \infty$ such that

$$\Gamma_{\text{hypo}}^\infty(\chi_R, \chi_R) \leq \frac{C_M}{R^2} \mathbf{1}_{B_{2R} \setminus B_R}.$$

Hence the second term converges to 0 in $L^1(\pi_\infty)$, while the first term converges by dominated convergence, using the pointwise bound above by the Π_{k-1}^∞ -average of $\Gamma_{\text{hypo},k}^\infty(F, F)$. Passing to the limit $R \rightarrow \infty$ yields

$$\text{Ent}_{\pi_\infty}(G_\varepsilon^2) \leq C_{\text{LSI}} \int_{\mathbb{R}^8} \Gamma_{\text{hypo}}^\infty(G_\varepsilon, G_\varepsilon) \, d\pi_\infty.$$

Integrating the previous pointwise estimate then gives

$$\text{Ent}_{\pi_\infty}(G_\varepsilon^2) \leq C_{\text{LSI}} \int_{(\mathbb{R}^8)^k} \Gamma_{\text{hypo},k}^\infty(F, F) \, d\Pi_k^\infty.$$

Since $G_\varepsilon^2 = m + \varepsilon$ decreases pointwise to $G^2 = m$ as $\varepsilon \downarrow 0$, and m is bounded while $t \mapsto t \log t$ is continuous on $[0, \infty)$, dominated convergence yields

$$\text{Ent}_{\pi_\infty}(G_\varepsilon^2) \rightarrow \text{Ent}_{\pi_\infty}(G^2).$$

Combining the bounds for the two entropy terms proves the claim for k . □

Remark 8.6 (Indicator-weighted thermodynamic observables). The thermodynamic link and plaquette observables of [?] are integrated against bounded indicators χ_t and $\chi_\square$ on the product spaces $(\mathbb{R}^8)^2$ and $(\mathbb{R}^8)^4$. Since

$$0 \leq \chi_t, \chi_\square \leq 1,$$

their integrability and entropy control are inherited from the supporting product measures $\pi_\infty^{\otimes 2}$ and $\pi_\infty^{\otimes 4}$ of Proposition ??; no additional hypothesis is required at this stage.

Remark 8.7 (Compatibility with thermodynamic product measures). Let $O \in \mathfrak{A}_{\text{loc}}^{\text{g.i.}}$ be a product of m generators from Definition ?. In the thermodynamic plaquette presentation of [?], an m -fold product of single-plaquette gauge-invariant observables is naturally evaluated on a $4m$ -point equilibrium product space. Proposition ?? therefore applies to the supporting product measure $\pi_\infty^{\otimes 4m}$, with the same logarithmic Sobolev inequality constant C_{LSI} as for π_∞ .

9 Discussion of excitations and propagators

The propagator on the finite regulator graph. Because the theory is finite, all correlation functions can be defined by source differentiation exactly. Their graph-theoretic support is determined by the finite regulator edge graph.

Definition 9.1 (Generating functional). *Introduce scalar sources $J = (J_x)_{x \in \mathcal{N}}$ and fermionic sources $\eta = (\eta_x)_{x \in \mathcal{N}}$, $\bar{\eta} = (\bar{\eta}_x)_{x \in \mathcal{N}}$. The generating functional is*

$$Z[J, \eta, \bar{\eta}] := \int \exp \left(-S + \sum_{x \in \mathcal{N}} J_x \phi_x + \sum_{x \in \mathcal{N}} \bar{\eta}_x \psi_x + \sum_{x \in \mathcal{N}} \bar{\psi}_x \eta_x \right) d\mu.$$

Definition 9.2 (Two-point functions). *Whenever $Z[0, 0, 0] \neq 0$, the scalar and fermion two-point functions are*

$$G_\phi(x, y) := \langle \phi_x \phi_y \rangle, \quad S_F(x, y)_{\alpha\beta} := \langle \psi_{x,\alpha} \bar{\psi}_{y,\beta} \rangle.$$

Proposition 9.3 (Background-field propagators). *Fix a gauge field U .*

1. *The scalar background covariance is*

$$G_\phi^{(U)} = (L_\eta + m_\phi^2 I)^{-1}.$$

2. *The fermion background covariance is*

$$S_F^{(U)} = M_F(U)^{-1}.$$

Proof. The scalar claim is the standard Gaussian covariance formula from Theorem ?. By Proposition ??, the matrix $M_F(U)$ is invertible for every U . The fermion claim then follows from differentiating the Berezin generating functional

$$\int e^{-\bar{\Psi} M_F(U) \Psi + \bar{\eta} \Psi + \bar{\Psi} \eta} d\bar{\Psi} d\Psi = \det M_F(U) e^{\bar{\eta} M_F(U)^{-1} \eta}.$$

□

Corollary 9.4 (Exact finite Euclidean functionals on the canonical local gauge-invariant sectors). *For every regulator size N , every $O \in \mathfrak{A}_{\text{loc}}^{\text{g.i.}}$, and every $\chi \in C_c(\mathbb{R}^4)$, the unnormalized Euclidean insertions*

$$\tilde{\omega}_N(\iota_N(O)) \quad \text{and} \quad \tilde{\omega}_N(j_N(\mathcal{W}_\chi))$$

are finite complex numbers.

Proof. By Proposition ??, each $\iota_N(O)$ is a bounded gauge-invariant bosonic observable. By Definition ??, each $j_N(\mathcal{W}_\chi)$ is a finite linear combination of bounded plaquette class functions, hence is again a bounded gauge-invariant bosonic observable. The finiteness of both unnormalized Euclidean functionals is therefore an immediate application of Theorem ??. $\square$

Theorem 9.5 (Exact finite bosonic reduction of gauge-invariant insertions). *Fix a rooted spanning forest $\mathcal{T}$ as in Theorem ??. Let $\mathcal{O} : \mathcal{U} \rightarrow \mathbb{C}$ be a bounded measurable gauge-invariant observable, viewed as an observable on $\mathcal{X}_{\mathcal{L}}$ by ignoring the color, fermion, and scalar variables. Then the unnormalized Euclidean functional satisfies*

$$\tilde{\omega}_N(\mathcal{O}) = Z_\phi \int_{\mathcal{U}} \mathcal{O}(U) e^{-S_W(U)} \det M_F(U) \, d\mu_{\mathcal{U}}(U),$$

where

$$Z_\phi = (2\pi)^{|\mathcal{N}|/2} \det(L_\eta + m_\phi^2 I)^{-1/2}.$$

Moreover, equivalently,

$$\tilde{\omega}_N(\mathcal{O}) = Z_\phi \int_{\mathcal{U}} \delta_{\mathcal{T}}(U) \mathcal{O}(U) e^{-S_W(U)} \det M_F(U) \, d\mu_{\mathcal{U}}(U),$$

and after integrating out the constrained tree links this becomes an ordinary finite-dimensional integral over the cotree edge variables $\mathcal{E}_{\text{co}} = \mathcal{E} \setminus \mathcal{T}$.

Proof. Because the color sector is normalized and decoupled from the action, the color integration contributes the unit factor. Since $\mathcal{O}$ depends only on the gauge field, (??) gives

$$\tilde{\omega}_N(\mathcal{O}) = \int_{\mathcal{U} \times \Lambda_F \times \mathcal{S}} \mathcal{O}(U) e^{-S_W(U) - S_F(U, \bar{\psi}, \psi) - S_\phi(\phi)} \, d\mu_{\mathcal{U}}(U) \, d\mu_F \, d\phi.$$

Apply Theorem ?? to integrate out the fermions:

$$\tilde{\omega}_N(\mathcal{O}) = \int_{\mathcal{U} \times \mathcal{S}} \mathcal{O}(U) e^{-S_W(U) - S_\phi(\phi)} \det M_F(U) \, d\mu_{\mathcal{U}}(U) \, d\phi.$$

Now apply Theorem ??. Since the scalar action is independent of U , the scalar integral factors out as Z_ϕ , yielding

$$\tilde{\omega}_N(\mathcal{O}) = Z_\phi \int_{\mathcal{U}} \mathcal{O}(U) e^{-S_W(U)} \det M_F(U) \, d\mu_{\mathcal{U}}(U).$$

By Proposition ??, the original finite action and product measure are gauge invariant. Since $\mathcal{O}$ is gauge invariant, the full integrand before the fermion and scalar integrations is gauge invariant. Hence the gauge-fixed representation from Corollary ?? applies, and because the Faddeev–Popov determinant is identically 1 by Theorem ??, one obtains

$$\tilde{\omega}_N(\mathcal{O}) = Z_\phi \int_{\mathcal{U}} \delta_{\mathcal{T}}(U) \mathcal{O}(U) e^{-S_W(U)} \det M_F(U) \, d\mu_{\mathcal{U}}(U).$$

The final statement is exactly the last sentence of Corollary ??. $\square$

Corollary 9.6 (Exact finite bosonic reduction on the canonical local gauge-invariant sectors). *For every regulator size N , every $O \in \mathfrak{A}_{\text{loc}}^{\text{g.i.}}$, and every $\chi \in C_c(\mathbb{R}^4)$,*

$$\begin{aligned}\tilde{\omega}_N(\iota_N(O)) &= Z_\phi \int_{\mathcal{U}} \delta_{\mathcal{T}}(U) \iota_N(O)(U) e^{-S_W(U)} \det M_F(U) \, d\mu_{\mathcal{U}}(U), \\ \tilde{\omega}_N(j_N(\mathcal{W}_\chi)) &= Z_\phi \int_{\mathcal{U}} \delta_{\mathcal{T}}(U) j_N(\mathcal{W}_\chi)(U) e^{-S_W(U)} \det M_F(U) \, d\mu_{\mathcal{U}}(U).\end{aligned}$$

In each case, after integrating out the constrained tree links, the right-hand side is an ordinary finite-dimensional integral over the cotree edge variables.

Proof. By Proposition ??, $\iota_N(O)$ is gauge invariant for every $O \in \mathfrak{A}_{\text{loc}}^{\text{g.i.}}$. By Definition ??, $j_N(\mathcal{W}_\chi)$ is a finite linear combination of gauge-invariant plaquette class functions and is therefore gauge invariant. Apply Theorem ?? to each observable. $\square$

Proposition 9.7 (Unconditional fermion invertibility). *For every gauge field $U \in \mathcal{U}$, the matrix $M_F(U)$ is invertible. More precisely, if*

$$M_F(U) = D - H(U), \quad D := \text{diag}(m_F + \text{indeg}(x))_{x \in \mathcal{N}} \otimes I_3,$$

then

$$\|D^{-1}H(U)\|_{\infty, 2 \rightarrow \infty, 2} \leq \max_{y \in \mathcal{N}} \frac{\text{indeg}(y)}{m_F + \text{indeg}(y)} < 1,$$

where

$$\|\xi\|_{\infty, 2} := \max_{x \in \mathcal{N}} \|\xi_x\|_2 \quad \text{for } \xi = (\xi_x)_{x \in \mathcal{N}} \in \bigoplus_{x \in \mathcal{N}} \mathbb{C}_x^3.$$

Proof. For $\xi = (\xi_x)_{x \in \mathcal{N}}$, the y -block of $H(U)\xi$ is

$$(H(U)\xi)_y = \sum_{\substack{e \in \mathcal{E} \\ t(e)=y}} U(e) \xi_{s(e)}.$$

Since each $U(e) \in SU(3)$ is unitary,

$$\|U(e)\xi_{s(e)}\|_2 = \|\xi_{s(e)}\|_2.$$

Therefore

$$\|(H(U)\xi)_y\|_2 \leq \sum_{\substack{e \in \mathcal{E} \\ t(e)=y}} \|\xi_{s(e)}\|_2 \leq \text{indeg}(y) \|\xi\|_{\infty, 2}.$$

Multiplying by D^{-1} gives

$$\|(D^{-1}H(U)\xi)_y\|_2 \leq \frac{\text{indeg}(y)}{m_F + \text{indeg}(y)} \|\xi\|_{\infty, 2}.$$

Taking the maximum over y yields the stated bound. Because $m_F > 0$, every fraction $\text{indeg}(y)/(m_F + \text{indeg}(y))$ is strictly less than 1. Hence $\|D^{-1}H(U)\|_{\infty, 2 \rightarrow \infty, 2} < 1$, so the Neumann series for $(I - D^{-1}H(U))^{-1}$ converges. Since

$$M_F(U) = D(I - D^{-1}H(U))$$

and D is invertible, $M_F(U)$ is invertible. $\square$

Proposition 9.8 (Fermion hopping expansion). *Let*

$$M_F(U) = D - H(U),$$

where

$$D := \text{diag}(m_F + \text{indeg}(x))_{x \in \mathcal{N}} \otimes I_3$$

and $H(U)$ is the off-diagonal hopping part from Proposition ?? . Then

$$M_F(U)^{-1} = \sum_{n=0}^{\infty} (D^{-1}H(U))^n D^{-1},$$

and the (x, y) matrix element is a convergent sum over directed paths $\gamma : y \rightsquigarrow x$ in $\mathcal{E}$. Therefore the support of the fermion propagator is controlled by the finite regulator graph, and its combinatorics are inherited from the deterministic regulator incidence of the companion data in [?].

Proof. By Proposition ?? ,

$$\|D^{-1}H(U)\|_{\infty, 2 \rightarrow \infty, 2} < 1.$$

Hence the Neumann series for $(I - D^{-1}H(U))^{-1}$ converges in that operator norm, giving

$$(I - D^{-1}H(U))^{-1} = \sum_{n=0}^{\infty} (D^{-1}H(U))^n.$$

Multiplying on the right by D^{-1} yields the displayed formula for $M_F(U)^{-1}$. Each power $(D^{-1}H(U))^n$ is a sum over length- n compositions of off-diagonal hopping operators, hence over directed paths of length n in the graph. The only admissible steps for those paths are edges in $\mathcal{E}$, as determined by the finite regulator data from [?]. $\square$

Theorem 9.9 (Absolutely convergent fermion loop expansion and exact effective gauge action). *For every gauge field $U \in \mathcal{U}$, define*

$$K(U) := D^{-1}H(U), \quad q_* := \max_{y \in \mathcal{N}} \frac{\text{indeg}(y)}{m_F + \text{indeg}(y)} < 1.$$

Then the following hold.

1. The spectrum of $K(U)$ lies in the closed disk $\{z \in \mathbb{C} : |z| \leq q_*\}$.
2. The principal matrix logarithm of $I - K(U)$ is well defined and satisfies

$$(I - K(U)) = - \sum_{n=1}^{\infty} \frac{K(U)^n}{n},$$

with absolute convergence in the operator norm $\|\cdot\|_{\infty, 2 \rightarrow \infty, 2}$.

3. The fermion determinant admits the exact logarithmic expansion

$$\log \det M_F(U) = \log \det D - \sum_{n=1}^{\infty} \frac{1}{n} (K(U)^n).$$

4. For each $n \geq 1$, let $\mathcal{L}_n^{\text{cl}}$ be the finite set of closed based directed walks

$$\ell = (e_1, \dots, e_n)$$

in $\mathcal{E}$, meaning

$$t(e_j) = s(e_{j+1}) \quad (1 \leq j \leq n-1), \quad t(e_n) = s(e_1).$$

For $\ell \in \mathcal{L}_n^{\text{cl}}$, define

$$w(\ell) := \prod_{j=1}^n \frac{1}{m_F + \text{indeg}(t(e_j))}, \quad U(\ell) := U(e_n) \cdots U(e_1).$$

Then

$$(K(U)^n) = \sum_{\ell \in \mathcal{L}_n^{\text{cl}}} w(\ell) \text{tr}(U(\ell)).$$

In particular, each $(K(U)^n)$ is gauge invariant.

5. The series is uniformly absolutely summable in the sense that

$$|(K(U)^n)| \leq 3|\mathcal{N}| q_*^n,$$

and therefore, for every $L \geq 0$,

$$\sum_{n=L+1}^{\infty} \frac{1}{n} |(K(U)^n)| \leq \frac{3|\mathcal{N}|}{1-q_*} q_*^{L+1}.$$

Proof. By Proposition ??,

$$\|K(U)\|_{\infty, 2 \rightarrow \infty, 2} \leq q_* < 1.$$

Hence the spectral radius satisfies

$$\rho(K(U)) \leq \|K(U)\|_{\infty, 2 \rightarrow \infty, 2} \leq q_*,$$

which proves the first claim.

Because $\|K(U)\|_{\infty, 2 \rightarrow \infty, 2} < 1$, the scalar power series

$$-\sum_{n=1}^{\infty} \frac{z^n}{n}$$

for $\log(1-z)$ converges absolutely on the closed disk $|z| \leq q_*$. Therefore the matrix series

$$L(U) := -\sum_{n=1}^{\infty} \frac{K(U)^n}{n}$$

converges absolutely in $\|\cdot\|_{\infty, 2 \rightarrow \infty, 2}$. Since $\sigma(K(U)) \subset \{|z| \leq q_*\}$, the spectrum of $I - K(U)$ is contained in the disk $\{w : |w - 1| \leq q_*\}$, which avoids the nonpositive real axis because $q_* < 1$. Thus the principal matrix logarithm is defined, and by the holomorphic functional calculus for the principal branch,

$$(I - K(U)) = L(U).$$

This proves the second claim.

Since

$$M_F(U) = D(I - K(U)),$$

we have

$$\det M_F(U) = \det D \cdot \det(I - K(U)).$$

The matrix D is diagonal with strictly positive real entries, so $\log \det D$ is the ordinary real logarithm. In finite dimension,

$$\det(I - K(U)) = \exp\left(\text{tr}(I - K(U))\right).$$

Taking logarithms and inserting the series from the second claim yields

$$\log \det M_F(U) = \log \det D + \text{tr}(I - K(U)) = \log \det D - \sum_{n=1}^{\infty} \frac{1}{n} \text{tr}(K(U)^n),$$

which proves the third claim.

To identify the trace terms, note that for $\xi = (\xi_x)_{x \in \mathcal{N}}$,

$$(K(U)\xi)_y = \frac{1}{m_F + \text{indeg}(y)} \sum_{\substack{e \in \mathcal{E} \\ t(e)=y}} U(e) \xi_{s(e)}.$$

Iterating this identity shows that the (x, x) -block of $K(U)^n$ is the sum over all directed length- n paths

$$x = x_0 \xrightarrow{e_1} x_1 \xrightarrow{e_2} \cdots \xrightarrow{e_n} x_n = x$$

of the matrices

$$\left(\prod_{j=1}^n \frac{1}{m_F + \text{indeg}(x_j)} \right) U(e_n) \cdots U(e_1).$$

Taking the 3×3 trace of the diagonal block and then summing over $x \in \mathcal{N}$ gives

$$(K(U)^n) = \sum_{\ell \in \mathcal{L}_n^{\text{cl}}} w(\ell) \text{tr}(U(\ell)),$$

where $\mathcal{L}_n^{\text{cl}}$ is precisely the set of closed based directed walks of length n . This proves the displayed loop expansion.

Under a gauge transformation $U(e) \mapsto \Omega(t(e))U(e)\Omega(s(e))^{-1}$, the product along a closed walk transforms by conjugation:

$$U(\ell) \mapsto \Omega(s(e_1))U(\ell)\Omega(s(e_1))^{-1}.$$

Hence $\text{tr}(U(\ell))$ is gauge invariant, and therefore so is $(K(U)^n)$.

Finally, for any matrix A on $\bigoplus_{x \in \mathcal{N}} \mathbb{C}_x^3$,

$$|A| \leq \sum_{x \in \mathcal{N}} |\text{tr} A_{xx}| \leq 3|\mathcal{N}| \|A\|_{\infty, 2 \rightarrow \infty, 2}.$$

Applying this with $A = K(U)^n$ gives

$$|(K(U)^n)| \leq 3|\mathcal{N}| \|K(U)\|_{\infty, 2 \rightarrow \infty, 2}^n \leq 3|\mathcal{N}| q_*^n.$$

Therefore

$$\sum_{n=L+1}^{\infty} \frac{1}{n} |(K(U)^n)| \leq 3|\mathcal{N}| \sum_{n=L+1}^{\infty} q_*^n = \frac{3|\mathcal{N}|}{1 - q_*} q_*^{L+1},$$

which proves the last claim. $\square$

Corollary 9.10 (Exact loop-effective bosonic action). *Define the exact fermionic effective gauge action by*

$$S_{\text{eff},F}(U) := \sum_{n=1}^{\infty} \frac{1}{n} (K(U)^n), \quad K(U) = D^{-1}H(U).$$

Then the series converges absolutely for every $U \in \mathcal{U}$, each partial sum is gauge invariant, and

$$\det M_F(U) = \det D e^{-S_{\text{eff},F}(U)}.$$

Consequently, for every bounded measurable gauge-invariant observable $\mathcal{O} : \mathcal{U} \rightarrow \mathbb{C}$,

$$\tilde{\omega}_N(\mathcal{O}) = Z_\phi \det D \int_{\mathcal{U}} \delta_{\mathcal{T}}(U) \mathcal{O}(U) e^{-S_W(U) - S_{\text{eff},F}(U)} d\mu_{\mathcal{U}}(U).$$

Proof. Absolute convergence and gauge invariance follow directly from Theorem ???. The determinant identity is the exponential form of the logarithmic expansion proved there:

$$\log \det M_F(U) = \log \det D - S_{\text{eff},F}(U).$$

Exponentiating gives

$$\det M_F(U) = \det D e^{-S_{\text{eff},F}(U)}.$$

Substituting this identity into Theorem ??? yields the stated bosonic reduction formula. $\square$

Wilson loops and the area law. The Wilson loop remains the basic gauge-invariant probe of confinement.

Definition 9.11 (Wilson loop expectation). *For an oriented closed loop γ in the lattice,*

$$W(\gamma) := \text{tr} \left(\prod_{e \in \gamma} U(e) \right), \quad \langle W(\gamma) \rangle := Z^{-1} \int W(\gamma) e^{-S} d\mu,$$

whenever $Z \neq 0$.

Remark 9.12 (Area-law criterion). If a family of loops γ admits spanning surfaces Σ_γ in the embedded spacetime geometry and one can prove

$$|\langle W(\gamma) \rangle| \leq C e^{-\sigma \text{Area}(\Sigma_\gamma)}$$

with $\sigma > 0$ independent of the loop size, then one has the usual Euclidean area-law signature of confinement [?]. The present paper does *not* establish such a bound; it merely provides the exact finite-lattice measure in which the question is well posed.

10 Conclusion

Summary of the quantum data package. Starting from the deterministic regulator gauge package of [?], this paper has completed the exact finite-dimensional quantization step. The link variables were promoted from data-derived transporters to independent $SU(3)$ variables, the full configuration space and product measure were defined, and the partition function was proved finite for the free massive scalar sector used here. The gauge redundancy was solved globally by a rooted maximal-tree gauge, with exact orbit uniqueness, exact Faddeev–Popov identity, constant determinant, and a decoupled ghost sector. The fermions integrate to a finite determinant, the scalar sector is an exact Gaussian field with stiffness operator $L_\eta + m_\phi^2 I$, and the continuum action $S_{\text{YM}}^{\text{cont}}$ (in the stationary formulation of [?]) is paired here with a canonical continuum-indexed class of local gauge-invariant plaquette observables and its finite-regulator approximation map.

Forward look to the thermodynamic theory. The $N \rightarrow \infty$ thermodynamic limit developed in the companion papers places the fixed- N partition functions constructed here into the stationary mean-field setting: [?] proves

$$\pi_{N,1} \Rightarrow \pi_\infty,$$

while [?] identifies the continuum limit action $S_{\text{YM}}^{\text{cont}}$ and associated thermodynamic observables on π_∞ . The paper supplies the exact finite-dimensional quantization step on the deterministic regulator lattice, together with the local Haar-coordinate analysis and the explicit gauge fixing needed to pass from data package $\mathcal{Q}^{(N)}$ to its quantum ensemble. In addition, it fixes the continuum-indexed gauge-invariant observable class that is local on each finite regulator, uniformly bounded on compact windows, nontrivial, and carried in the thermodynamic presentation by the equilibrium product measures $\pi_\infty^{\otimes k}$ with the transferred logarithmic Sobolev control.